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Introduction:

This report presents the design and SPICE verification of an open-loop voltage amplifier, which is constructed entirely from MOSFET devices. Operating from a single 3.3 V rail, it is designed to meet specification requirements relating to its voltage gain, bandwidth, input resistance, output swing, load sensitivity, power consumption, and the branch current. To achieve a high overall gain, the circuit uses multiple cascaded amplification stages with resistive biasing and AC coupling between the stages. Additionally, a source follower output stage is included to enhance the load driving ability and reduce gain loss when connected to that specific load.

Objective:

The goal of this project is to design and evaluate a voltage amplifier built exclusively from MOSFETs that satisfies the specified design criteria. This is achieved by carefully choosing the circuit components, determining appropriate transistor dimensions, and implementing effective biasing networks. The design process involves performing hand calculations to predict the amplifier's theoretical behaviour, then using KiCad SPICE simulations to validate the circuit. Essential performance metrics include open-loop gain, -3 dB bandwidth, input resistance, output swing, load sensitivity, total DC power consumption, and maximum branch current. In addition, the project aims to assess the accuracy of the design by comparing the analytical results with the simulation data to determine how well the final circuit aligns with the expected device performance.

Design Process:

The design started by choosing a circuit structure which could achieve a high open-loop gain while operating from a 3.3V supply, while also remaining within the specified power and current limits. A multistage MOSFET amplifier was selected for this purpose. The circuit consists of three common-source stages to provide the majority of the voltage gain, followed by a common drain (source follower) stage to improve the ability to drive the load and minimize the gain reduction at the output. The next step involved establishing appropriate DC biasing conditions for each stage. For the first 3 stages, voltage dividers biasing was used to set the gate voltage to approximately 0.825V. High-value resistors, which fall in the Mega Ohm range, were chosen to keep the bias currents very low, reducing the overall power consumption while maintaining a large input resistance. Each common-source stage used a $30k\Omega$ drain resistor, which allowed for sufficient voltage drops across the transistor while maintaining a reasonable drain current for amplification. Capacitors were inserted between stages to enable AC signal transfer while isolating the DC biasing of each stage.

Once the gain stages were defined, the output stage was implemented as a source follower. A separate bias network was used to set a higher gate voltage for this stage so that the source voltage would be at a suitable level to drive the load. The source transistor was selected to control the current through the output transistor while ensuring the bench current remained within limits. An output coupling capacitor was included to block DC and allow only the amplified AC signal to reach the load. The load resistor and capacitor were used to model the required output conditions. After selecting the circuit configuration and component values, hand calculations were carried out to estimate key parameters such as node voltages, drain currents, transconductances, individual stage gains, overall gain, input resistance, output swing, and total power consumption. These calculations provided an initial reference for the design. The circuit was then built in KiCad and tested using DC, AC, and transient analyses. The DC operating point analyses verified correct biasing and current levels, the AC analysis was used to evaluate gain, bandwidth, and input resistance, and the transient analysis allowed observation of time domain behaviour and output swing. Based on these results, adjustments were made until the amplifier satisfied all required specifications.

Circuit Under Test:

The circuit being analyzed is a four-stage MOSFET voltage amplifier, which is powered by a single 3.3 V supply. The input signal is introduced through source V1, which includes a 50Ωsource resistance, and is AC-coupled into the first stage using capacitor C1. The first 3 stages use transistors Q1, Q2, and Q3 configured as common source amplifiers, which are responsible for generating the majority of the voltage gain. Each stage is biased using a resistor-divider network (R6-R7, R8-R9, R10-R11) to establish the appropriate gate DC voltage. The drain resistors R1, R2, and R3 serve as the load for each stage. Capacitors C2, C3, and C4 are placed between the stages to allow the AC signal to pass while isolating the DC bias conditions of each stage. The final stage, built around transistor Q4, is a common-drain (source follower) configuration that acts as an output buffer. Its gate voltage is set using resistors R12 and R13, while resistor R14 controls the source current. This stage helps reduce the output resistance and improves the circuit's ability to drive the connected load. The output is AC-coupled through capacitor C5, ensuring that only amplified signals reach the load while blocking DC components. The load itself is modeled using R15 and C6. In summary, the circuit consists of three cascaded common-source stages followed by a source follower output stage, which provides high overall gain along with improved load driving performance.

Experimental Results:

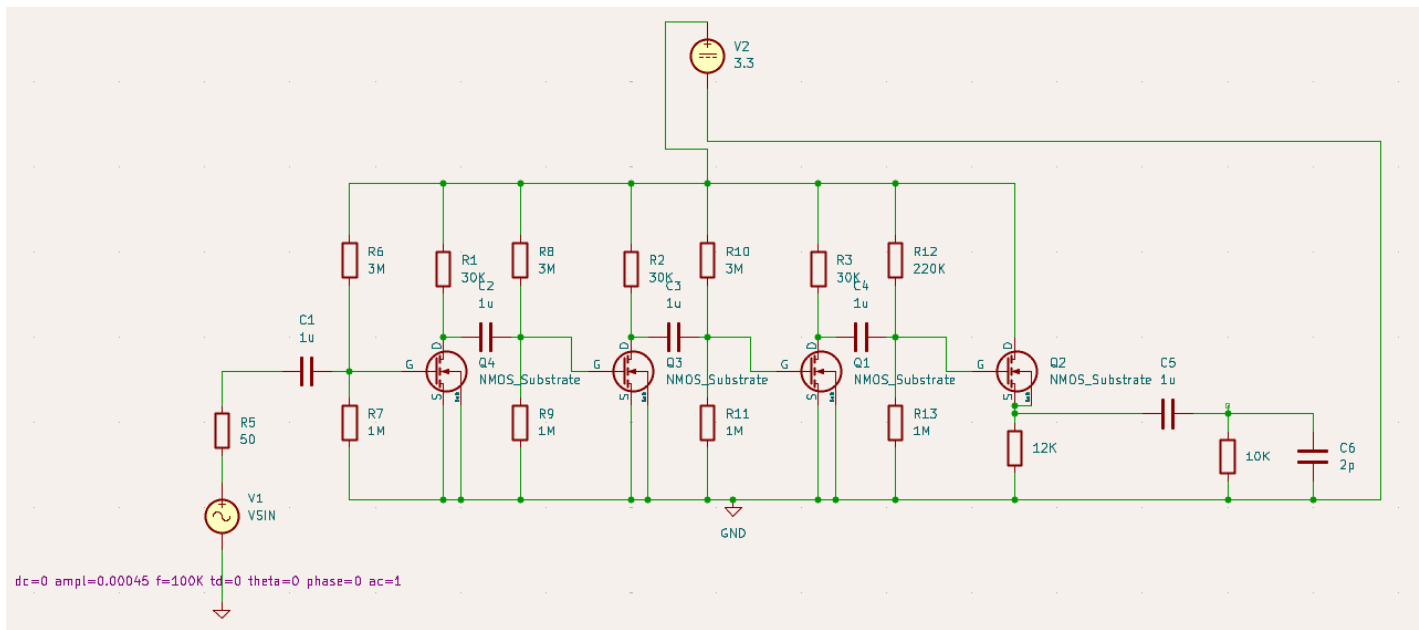


Figure 1. Schematic of the MOSFET Amplifier Built in KiCad.

Experimental Results:

STAGE 1, 2, 3 ANALYSIS

Stage 1, 2, 3 DC Analysis		
Gate Bias Voltage	Overdrive Voltage	Drain Current

Stage 1, 2, 3 DC Analysis		
$V_{B1} = V_{DD} \frac{R_7}{R_6 + R_7}$ $V_{B1} = 3.3 \times \frac{1M}{3M + 1M}$ $V_{B1} = 3.3 \times \frac{1}{4}$ $V_{B1} = 0.825 V$	$V_{S1} = 0 V$ $V_{GS1} = V_{G1} = 0.825 V$ $V_{OV1} = V_{GS1} - V_{tn}$ $V_{OV1} = 0.825 - 0.63$ $V_{OV1} = 0.195 V$	$I_D = \frac{1}{2} k_n \left(\frac{W}{L} \right)_1 (V_{OV})^2$ $I_{D1} = \frac{1}{2} (100 \mu A / V^2) (22) (0.195)^2$ $I_{D1} = 41.83 \mu A$

Stage 1, 2, 3 AC Analysis			
Output Resistance of the transistor	Transconductance	Input Resistance	Output Node Resistance
$r_{o1} = \frac{1}{\lambda \cdot I_{D1}}$ $r_{o1} = \frac{1}{(0.04) \cdot (41.83 \mu A)}$ $r_{o1} = 597.7 k\Omega$	$g_{m1} = \frac{2 \cdot I_{D1}}{V_{OV1}}$ $g_{m1} = \frac{2 \cdot 41.83 \mu A}{0.195 V}$ $g_{m1} = 0.429 mS$	$R_{in2} = R_6 R_7$ $R_{in2} = 3M 1M$ $R_{in2} = 750 k\Omega$	$R_{o,1} = R_1 r_{o1} R_{in2}$ $R_{o,1} = 30 k\Omega 597.7 k\Omega 750 k\Omega$ $R_{o,1} = 27.62 k\Omega$

Stage 1 Gain:

$$A_{v1} = - g_{m1} \cdot R_{o,1}$$

$$A_{v1} = - (0.429 mS)(27.62 k\Omega)$$

$$A_{v1} = - 11.85 V/V$$

$$|A_{v1}| = 11.85 V/V$$

$$A_{v1, dB} = 20 \log_{10}(11.85)$$

$$A_{v1, dB} = 21.48 dB$$

Gate Capacitance:

$$C_{GS1} = \frac{2}{3} C_{ox} \cdot W \cdot L$$

$$C_{GS1} = \frac{2}{3} (4.3 fF / \mu m^2) (22) (1)$$

$$C_{GS1} = 63.07 fF$$

Drain Capacitance:

$$C_{GD1} = C_{ov} \cdot W$$

$$C_{GD1} = (0.4fF/\mu m)(22)$$

$$C_{GD1} = 8.8 fF$$

Stage 1 Bandwidth:

$$C_{GS2} = 63.07 fF$$

$$C_{node1} \cong C_{GD1} + C_{GS2}$$

$$C_{node1} \cong 8.8 + 63.07$$

$$C_{node1} \cong 71.87 fF$$

$$f_{p1} = \frac{1}{2\pi \cdot R_{node} \cdot C_{node}}$$

$$f_{p1} = \frac{1}{2\pi \cdot (27.62k\Omega) \cdot (71.87 fF)}$$

$$f_{p1} = 80.18 MHz$$

Stage 2: $V_{B2} = 0.825 V$ $V_{OV2} = 0.195 V$ $I_{D2} = 41.83\mu A$ $r_{o2} = 597.7 k\Omega$ $g_{m2} = 0.429 mS$ $R_{o,2} = 27.62k\Omega$ $A_{v2} = -11.85 V/V$ $A_{v2,dB} = 21.48 dB$ $C_{GS2} = 63.07 fF$ $C_{GD2} = 8.8 fF$ $f_{p2} = 80.18 MHz$	Stage 3: $V_{B3} = 0.825 V$ $V_{OV3} = 0.195 V$ $I_{D3} = 41.83\mu A$ $r_{o3} = 597.7 k\Omega$ $g_{m3} = 0.429 mS$ $R_{o,3} = 27.62k\Omega$ $A_{v3} = -11.85 V/V$ $A_{v3,dB} = 21.48 dB$ $C_{GS3} = 63.07 fF$ $C_{GD3} = 8.8 fF$ $f_{p3} = 80.18 MHz$
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All values are the same for stages 1, 2, and 3

STAGE 4 ANALYSIS

Stage 4 DC Analysis		
Gate Bias Voltage	Source Voltage and Drain Current	Overdrive Voltage and Saturation Check
$V_{B4} = V_{G4} = V_{DD} \frac{R_{13}}{R_{12} + R_{13}}$ $V_{B4} = 3.3 \times \frac{1M}{220k + 1M}$ $V_{B4} = 3.3 \times \frac{1}{1.22}$ $V_{B4} = 2.705 V$	$V_{S4} = I_{D4} R_{13}$ $V_{OV4} = V_{GS4} - V_{tn} = (V_{G4} - V_{S4}) - V_{tn}$ $I_D = \frac{1}{2} k_n \left(\frac{W}{L}\right)_1 (V_{OV})^2$	$V_{OV4} = 2.075 - 1.896$ $V_{OV4} = 0.179 V$ $V_{D4} = 3.3 V$

Stage 4 DC Analysis		
	$I_{D4} = \frac{1}{2}(100\mu A/V^2)(100)(V_{OV4})^2$ $I_{D4} = 5mA/V^2(V_{OV4})^2$ $V_{OV4} = V_{G4} - V_{S4} - V_{tn}$ $V_{OV4} = 2.705 - V_{S4} - 0.63$ $V_{OV4} = 2.075 - V_{S4}$ $I_{D4} = \frac{V_{S4}}{12k}$ $\frac{V_{S4}}{12k} = 5mA(2.075 - V_{S4})^2$ $V_{S4} = 1.896V$ $I_{D4} = \frac{1.896V}{12k}$ $I_{D4} = 158\mu A$	$V_{DS4} = V_{D4} - V_{S4}$ $V_{DS4} = 3.3 - 1.896$ $V_{DS4} = 1.404V$ $V_{DS4} \geq V_{OV4}$ $1.404 \geq 0.179$ Stauration Mode

Stage 4 AC Analysis		
Output Resistance of the transistor	Transconductance	Input Resistance
$r_{o4} = \frac{1}{\lambda \cdot I_{D4}}$ $r_{o4} = \frac{1}{(0.04) \cdot (158\mu A)}$ $r_{o4} = 158.2k$	$g_{m4} = \frac{2 \cdot I_{D4}}{V_{OV4}}$ $g_{m1} = \frac{2 \cdot 158\mu A}{0.179V}$ $g_{m4} = 1.765mS$	$R_{in4} = R_{12} R_{13}$ $R_{in4} = 220k 1M$ $R_{in4} = 180.3k$

Stage 4 Gains		
Loaded Gain	Unloaded Gain	Load Sensitivity
$A_{v4} = \frac{g_m \cdot R'_S}{1 + g_m \cdot R'_S}$ $R'_S = R_{14} r_{o4} R_{15}$ $R'_S = 12k 158.2k 10k$ $R'_S = 5.27k$ $A_{v4} = \frac{(1.765mS) \cdot (5.27k\Omega)}{1 + (1.765mS) \cdot (5.27k\Omega)}$ $A_{v4} = \frac{9.3}{10.3}$	$R'_S = R_{14} r_{o4}$ $R'_S = 12k 158.2k$ $R'_S = 11.16k$ $A_{v4, unloaded} = \frac{g_m \cdot R'_S}{1 + g_m \cdot R'_S}$ $A_{v4} = \frac{(1.765mS) \cdot (11.16k\Omega)}{1 + (1.765mS) \cdot (11.16k\Omega)}$	$\% gain reduction = \left \frac{A_{v, unloaded} - A_{v, loaded}}{A_{v, unloaded}} \right \times 100\%$ $\% gain reduction = \left \frac{0.952 - 0.903}{0.952} \right \times 100$ $\% gain reduction = 5.15\%$

$A_{v4} = 0.903 \text{ V/V}$ $A_{v4, dB} = 20\log_{10}(0.903)$ $A_{v4, dB} = -0.89 \text{ dB}$	$A_{v4} = \frac{19.7}{20.7}$ $A_{v4} = 0.952 \text{ V/V}$ $A_{v4, dB} = 20\log_{10}(0.952)$ $A_{v4, dB} = -0.43 \text{ dB}$	
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Stage 4 Capacitances		
Gate Capacitance	Drain Capacitance	Stage 4 Bandwidth Estimate
$C_{GS4} = \frac{2}{3}C_{ox} \cdot W \cdot L$ $C_{GS4} = \frac{2}{3}(4.3 \text{ fF}/\mu\text{m}^2)(100)(1)$ $C_{GS4} = 286.7 \text{ fF}$	$C_{GD4} = C_{ov} \cdot W$ $C_{GD4} = (0.4 \text{ fF}/\mu\text{m})(100)$ $C_{GD4} = 40 \text{ fF}$	$C_L = 2 \text{ pF}$ $R_{node4} = (\frac{1}{g_{m4}}) R_{14} r_{o4} R_{15}$ $\frac{1}{g_{m4}} = \frac{1}{1.765 \text{ mS}} = 567 \Omega$ $R_{node4} = 567 \Omega 12 \text{ k}\Omega 158.2 \text{ k}\Omega 10 \text{ k}\Omega$ $R_{node4} = 509 \Omega$ $f_{p1} = \frac{1}{2\pi \cdot R_{node4} \cdot C_{node4}}$ $f_{p1} = \frac{1}{2\pi \cdot (509) \cdot (2 \text{ pF})}$ $f_{p1} = 156.3 \text{ MHz}$

Total Amplifier		
First bias divider	Cascaded Gain of Stages 1–3	Overall Total Gain (with buffer)
$R_{top} = 3 \text{ M}$ $R_{bottom} = 1 \text{ M}$ $R_{in} = 3 \text{ M} 1 \text{ M}$ $R_{in} = \frac{(3)(1)}{3+1} \text{ M} = 0.75 \text{ M} = 750 \text{ k}$	$A_{v1} = -g_m \cdot R'_D$ $A_{v1} = -(0.429 \text{ mS})(27.62 \text{ k})$ $A_{v1} = A_{v2} = A_{v3} = -11.85 \text{ V/V}$ $A_v = 20\log_{10}(11.85)$ $A_v = 21.48 \text{ dB per stage}$ $A_{v,123} = (-11.85)^3 = -1664 \text{ V/V}$ $20\log_{10}(1664) = 64.43 \text{ dB}$	$A_{v,total} = A_{v,123} \cdot A_{v,4}$ $A_{v,total} = -1664 \cdot 0.906 = -1508 \text{ V/V}$ $20\log_{10}(1508) = 63.57 \text{ dB}$ $V_{out,pp} = A_{v,total} \cdot V_{in,pp}$ $V_{out,pp} = 1508 \cdot 0.9 \text{ mV} = 1.357 \text{ Vpp}$

Stage 1 drain branch: 41.83 μA

Stage 2 drain branch: 41.83 μA
 Stage 3 drain branch: 41.83 μA
 Stage 4 drain branch: 158.0 μA

Divider current: three 3M/1M dividers = $3(0.825) = 2.475 \text{ uA}$
 One 220k/1M divider: $3.3/1.22 \text{ M} = 2.705 \text{ uA}$

$$I_{DD,total} = 41.83 + 41.83 + 41.83 + 158 + 2.475 + 2.705$$

$$I_{DD,total} = 288.7 \mu\text{A}$$

$$P_{DC} = V_{DD} I_{DD,total} = 3.3(288.7 \mu\text{A}) = 952 \mu\text{W}$$

Theoretical DC power is about 0.95 mW.

Maximum Branch Current: $I_{branch,max} = I_{D4} = 158 \mu\text{A}$

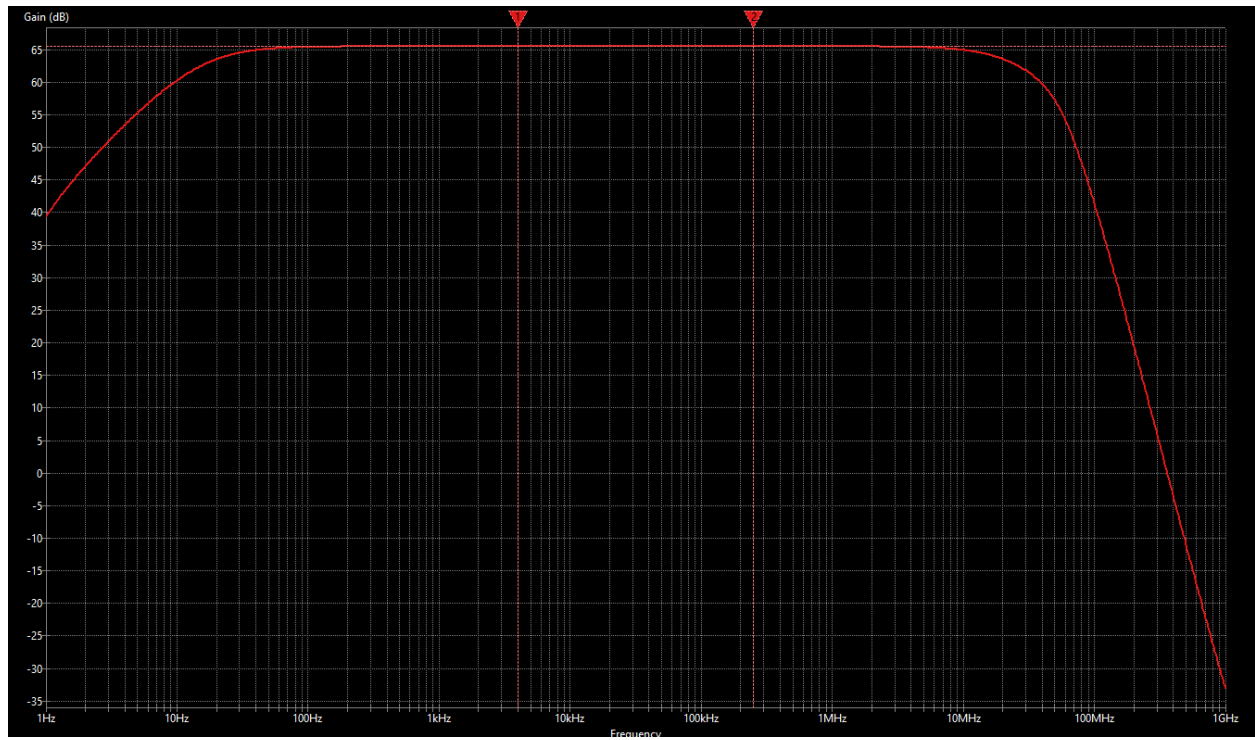
Performance Summary

Quantity	Theoretical Value	Simulated Value	Percentage Error (%)	Specification Satisfied
Input resistance (k Ω)	750	794	5.9%	Yes
Gate bias, stages 1–3 (V)	0.825 V	0.825 V	0.00%	Yes
Divider current, stages 1–3 (μA)	0.825 μA	—	—	Yes
Drain current, stages 1–3 (μA)	41.83 μA	46.11 μA	10.2%	Yes
Transconductance g_m , stages 1–3 (mS)	0.429 mS	0.473 mS	10.3%	Yes
Gain per CS stage (V/V)	-12.8 V/V	—	—	Yes
Gate bias, stage 4 (V)	2.705 V	2.705 V	0.00%	Yes
Source voltage, stage 4 (V)	1.896 V	1.90 V	0.21%	Yes
Drain/source current, stage 4 (μA)	158 μA	158.687 μA	0.43%	Yes
Stage 4 gain (V/V)	0.903 V/V	—	—	Yes
Total gain (V/V)	65.6 V/V	65.5 V/V	0.15%	Yes
Output swing (V)	1.72 V	1.55 V	9.9%	Yes
Total DC power (mW)	0.987 mW	0.997 mW	1.0%	Yes

Simulations:
OP ANALYSIS:

Node / Parameter	Simulated Value
Gate voltage, stages 1–3	0.825 V
Drain voltage, stages 1–3	1.917 V
Drain current, stages 1–3	46.113 μ A
Gate voltage, stage 4	2.705 V
Source voltage, stage 4	1.904 V
Drain/source current, stage 4	158.687 μ A
Supply current	302.205 μ A
Total DC power	0.997 mW

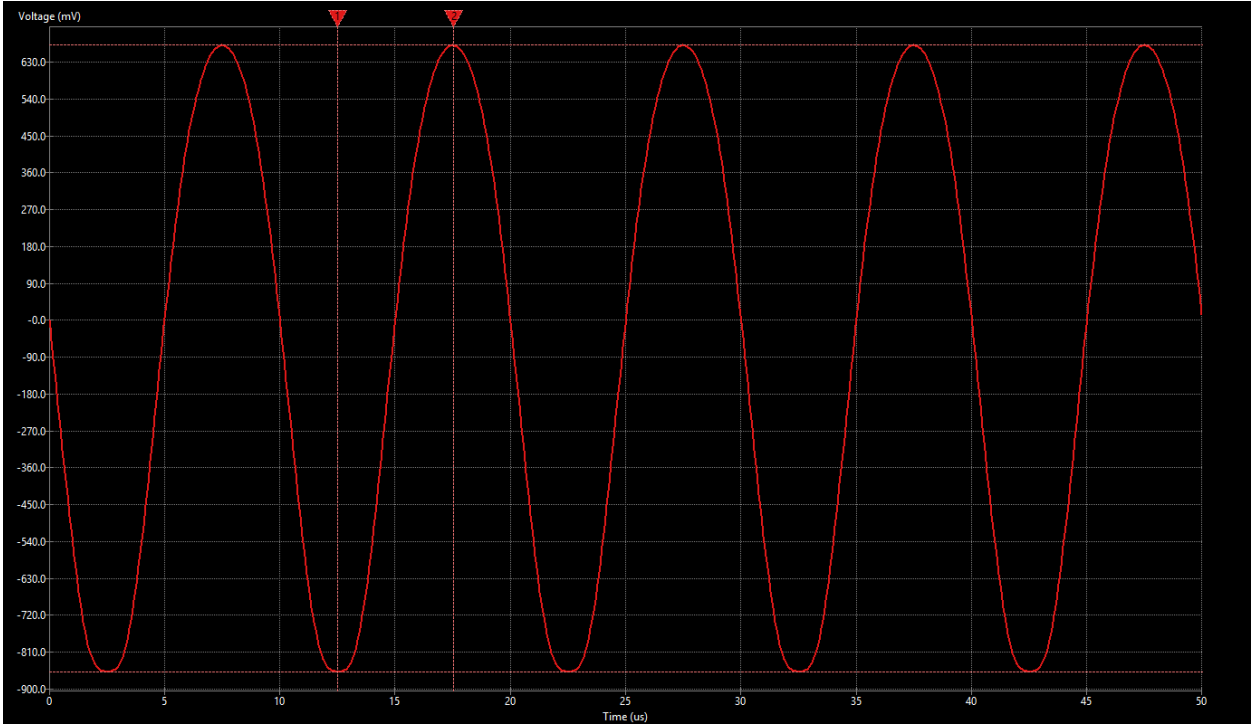
AC ANALYSIS(LOADED, R=10K):



AC Analysis(UNLOADED, R=100M):



Transient Analysis:



Conclusion and Remarks:

The MOSFET amplifier designed in this project was able to meet the key design requirements by using a combination of three common-source stages for gain and a source follower stage at the output for buffering. Simulation results closely matched the values obtained from hand calculations, with only minor percentage differences. This indicates that the theoretical analysis was a good approximation of the circuit's actual behavior. Any small discrepancies can be explained by non-ideal transistor characteristics, simplifications in the models, and assumptions made during manual calculations. This project also reinforced key concepts such as MOSFET biasing techniques, the design of multistage amplifiers, and the role of AC coupling between stages. It highlighted important design trade-offs, including those between gain, bandwidth, loading effects, and power consumption. In addition, the use of simulation proved to be essential for validating design choices and fine-tuning performance. Overall, the project demonstrated that proper biasing and careful selection of circuit stages are critical for meeting amplifier specifications while ensuring reliable operation.